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Sum = 0 + 2 + 4 = 6

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 0 & 4-\lambda \end{vmatrix} = \lambda^2 - 6\lambda + 8 = 0$$

$\lambda_1 = 4 \quad \lambda_2 = 2$

Trace $= \lambda_1 + \lambda_2 = 6$ ✓

$$\lambda_1 = 4 \rightarrow \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 2 \rightarrow \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Eig values $= \lambda_1 = 4$
$\lambda_2 = 2$

Eig vector $= x_1 = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$

$$x_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$